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Hepatitis B and C virus infections among potential blood donors in the Tahoua region, Niger

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ABSTRACT

Notwithstanding the advantages of blood transfusion, it exposes the recipient to the risk of transmission of numerous infectious agents, such as HBV and HCV. This study investigates the prevalence and risk factors of transfusion-transmissible HBV and HCV infections among blood donors in Tahoua, Niger. This study was conducted as a cross-sectional study at the Tahoua Regional Blood Transfusion Center (CRTS-TA). For each potential blood donor, HBV and HCV screening tests were performed using rapid diagnostic tests. A total of 3913 blood donors were included in the study. Blood donors were volunteers in 64.9% (n = 2538), and the majority (n = 2619; 66.9%) were regular blood donors. Blood group O Rhesus D-positive donors were the most represented, at 43.2% (n = 1691). The study reports a seroprevalence of HBV and HCV among blood donors of 7.2% (281/3913) and 1.3% (52/3913), respectively. On multivariate logistic analysis, the risk factors significantly associated with HBV infection were male blood donors (adjusted OR =2.25; IC95%: 1.27-4.16; p =0.008), family blood donors (adjusted OR =1.64; CI95%: 1.17-2.28; p =0.004) and new blood donors (adjusted OR =2.91; CI95%: 2.14-3.97; p <0.001). Risk factors significantly associated with HCV infection consisted solely of syphilis-positive blood donors (adjusted OR = 3.54; 95% CI: 1.46-7.63; p = 0.002).

The seroprevalence of HBV and HCV among blood donors at the CRTS-TA was at an intermediate endemicity and low endemicity, respectively, per the WHO classification. These findings underscore the need to reinforce strategies aimed at improving transfusion safety in Niger.

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Introduction

Hepatitis B and C viruses are major global health concerns, particularly in regions with limited transfusion safety infrastructure (Lelie et al., 2021). Blood transfusion is a medical procedure that saves lives and improves the health of many patients for whom it is

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indicated (Storch et al., 2019). In low-income countries, particularly in Africa, many patients requiring transfusions face two significant challenges: blood shortages and concerns about transfusion safety (Dei-Adomakoh et al., 2021). In sub-Saharan Africa, transfusion safety remains a considerable challenge, due to insufficient qualified human, financial, and material resources on the one hand, and the high prevalence of transmissible infections in the general population on the other (Dei-Adomakoh et al., 2021; Mpimbaza, 2020; Uyoga et al., 2023).

Notwithstanding the advantages of blood transfusion, it exposes the recipient to the risk of transmitting numerous infectious agents such as hepatitis B and C viruses (HBV and HCV) (Candotti et al., 2021; Gebreyes et al., 2025; Mangala et al., 2024; Puerto-Meredith et al., n.d.). The WHO therefore recommends screening for all blood donations. The world's highest prevalence rates of transfusion-transmissible infections are found on the African continent, with figures ranging from 4 to 25% for HBV, 0.5 to 8% for HCV, and 0.5 to 1% for HIV (Quintas et al., 2024).

In Niger, studies on the seroprevalence of markers for these infections in blood donors were conducted in Niamey and Zinder (Doutchi et al., 2021; Mayaki et al., 2013). In Niamey, Mayaki Z. et al. (Mayaki et al., 2013) reported a seroprevalence of 15.4% and 1.2% for HBsAg and HCV, respectively (Mayaki et al., 2013).

The CRTS-TA is one of the country's five major transfusion centers, with 10,853 blood bags collected in 2020, for a population estimated at 4.5 million inhabitants during the same period.

Despite previous studies in Niamey and Zinder, no data have been published from the Tahoua region since the CRTS-TA's establishment in 2010. This study addresses a critical gap by providing the first regional analysis of HBV and HCV seroprevalence and associated donor risk factors, contributing to national transfusion safety strategies.

Materials and Methods

Study setting

The study took place at the Tahoua Regional Blood Transfusion Center (CRTS-TA). Tahoua is one of the seven regions of Niger, located in the southwestern part of the country. The capital of the region is the city of Tahoua, which is also the fourth largest town in Niger. The region covers an area of approximately 106,677 km². It is bordered by the Agadez region to the North, the Zinder region to the East, and Nigeria to the South. Founded in the early 20th century, Tahoua became an important center during the French colonial period, serving as a hub for trade and administration. The region is characterized by a mix of traditional cultures and modern developments, with its landscapes ranging from arid deserts to fertile valleys. It serves as a crucial market town for traders, connecting the Sahel and Sahara regions.

Design and period study

This was a cross-sectional study of data collected from blood donation records at the CRTS-TA. The study was carried out at the CRTS-TA from January 15 to June 30, 2024.

Study population

The study participants were potential blood donors presenting at the blood bank of the CRTS-TA in Niger.

Sampling

Non-probabilistic sampling was performed. It consisted of a consecutive and exhaustive sample of all voluntary and family blood donors registered in the CRTS-TA laboratory register from January 15, 2024, to June 30, 2024.

Inclusion criteria

The selection of blood donors was based on eligibility criteria, in line with transfusion safety rules: an age between 18 and 65 years; a weight of greater than or equal to 50 kg; and compliance with a 3-month waiting period for men and a 4-month waiting period for women after a previous blood donation.

Exclusion criteria

Patients for whom the results of blood transfusion marker assays were not available were excluded from the study.

Data collection

Data were collected using a questionnaire that included socio-demographic data (age, sex, occupation, origin, marital status, type of donation, and donor status), as well as clinical data and results of HBV and HCV screening tests.

Biological analyses of specimens

Blood was collected by venipuncture at the elbow using the Vacutainer system (Becton, Dickinson UK Limited). Approximately 2 mL of whole blood was collected from each donor on a dry tube with a coagulation activator. Samples were kept at laboratory temperature for 5 minutes before being centrifuged at 1500 g for 5 minutes.

For each donor, tests for HBV and HCV were performed using HBsAg and anti-HCV SD BIOLINE tests (Abbott

Laboratories, Chicago, IL) according to the manufacturer's recommendations. These tests use the principle of immunochromatography for the qualitative detection of antibodies to these viruses in human serum, plasma, or whole blood. The sensitivity and specificity were 100% and 97.1% for the SD BIOLINE HBV test, and 100% and 98.1% for the SB BIOLINE HCV test, respectively (Kabamba et al., 2020).

Data analyses

Data were entered in Excel and analyzed using RStudio version 4.4.1. The Chi2 test (or Fisher's exact test) was used to compare categorical variables. Univariate and multivariate logistic analyses were performed to determine the risk factors associated with HBV and HCV carriage. Odds ratios and 95% confidence intervals were calculated. A p-value of less than 0.05 was considered statistically significant. Data were presented in tables.

Definition of terms

- **Family blood donor:** a donor who gives blood at the request of a member of the patient's family or community.
- **Voluntary blood donor:** a donor who gives blood of their own free will, without being forced to do so or receiving payment in exchange.
- **Regular blood donor**: is defined as an individual who donates blood consistently, typically at scheduled intervals, to support blood banks and hospitals in meeting the needs of patients.
- **New blood donor**: A person who is donating blood for the first time or who has not donated for at least 1 year.

Results

Basic characteristics of patients

A total of 3913 blood donors were included in this study. The mean age of the donors was 34.12 ± 10 years, with ages ranging from 17 to 64 years. The 25-34 age group was the most represented, comprising 36.6% (n = 1433). The majority of donors were male (sex ratio = 4.3), married (n = 2745; 70.2%), and self-employed (n = 1140; 29.1%). Blood donors were volunteers in 64.9% (n = 2538), and the majority were regular (n = 2619; 66.9%). Blood group O Rhesus D-positive donors were the most represented in this study, accounting for 43.2% (n = 1691), followed by blood group B Rhesus D-positive donors at 24.3% (n = 951) (Table 1).

Table 1 Basic characteristics of included patients

Variables	Charact.	Freq. (N)	Per. (%)	95%CI
Gender	Female	735	18.8	17.6 – 20.0
	Male	3178	81.2	80.0 – 82.4
Age (year)	(0 - 40]	2885	73.8	72.3 – 75.1
	(40 - 64]	1025	26.2	24.9 – 27.6
Occupation	Trader	1140	29.1	27.7 – 30.6
	Student	507	13.0	11.9 – 14.0
	Military	115	2.9	2.4 – 3.5
	Employed	251	6.4	5.6 – 7.2
	Unemployed	523	13.4	12.3 – 14.4
	Others*	1377	35.2	33.7 – 36.7
Marital status	Single	1163	29.7	28.3 – 31.2
	Married	2744	70.1	68.7 – 71.6
	Others$	6	0.2	0.0 – 0.3
Residence	Rural	694	17.7	16.5 – 18.9
	Urban	3219	82.3	81.1 – 83.5
Blood donor type	Family	1375	35.1	33.6 – 36.6
	Voluntary	2538	64.9	63.4 – 66.4
Blood donor status	New	1294	33.1	31.6 – 34.5
	Regular	2619	66.9	65.5 – 68.4
Blood group	A	904	23.1	21.8 – 24.4
	AB	186	4.8	4.1 – 5.4
	B	1012	25.9	24.5 – 27.2
	O	1811	46.3	44.7 – 47.8
D Rhesus	Negative	247	6.3	5.6 – 7.1
	Positive	3666	93.7	92.9 – 94.4

*farmers, stockbreeders, taximen, tailors, hairdressers, Koranic masters, and transporters; $Separated and Widowed; 95%CI: 95% = confidence interval; Charact.= Characteristics; Freq. = Frequency; Per. = Percentage.

Seroprevalence

The overall seroprevalence of HBV and HCV among blood donors at the CRTS-TA was 7.2% (281/3913) and 1.3% (52/3913), respectively. Among voluntary blood donors, the seroprevalence was 1.1% for HCV and 4.4% for HBV. Among family blood donors, the seroprevalence of HCV was 1.8% and of HBV 12.4%.

HBV and HCV-associated risk factors

For HBV, in the univariate logistic analysis, male donors with “other” occupations (farmers, stockbreeders, taximen, tailors, hairdressers, Koranic masters, transporters) from urban areas, family blood donors, new blood donors, HIV-positive donors, and syphilis-positive donors were significantly associated with a high risk of being HBV-positive. In multivariate logistic analysis,

only male donors, family blood donors, and new blood donors were significantly related to the risk of being HBV positive (Table 2).

For HCV, in univariate logistic analysis, blood donors from urban areas, new blood donors, and syphilis-positive donors were significantly associated with a high risk of being HCV-positive. In the multivariate logistic analysis, only syphilis-positive blood donors were significantly related to the risk of being HCV-positive (Table 3).

Table 2 Risk factors associated with HBV infection in blood donors

Variables		HBV		Univariate analysis			Multivariate analysis		
		Negative	Positive	OR	95%CI	*p*-value	aOR	95%CI	*p*-value
Gender	Female	703 (19.4)	32 (11.4)	1	-	-	1	-	-
	Male	2929 (80.6)	249 (88.6)	1.87	1.3-2.8	0.001	2.25	1.3-4.2	**0.008**
Age	(40 - 64]	958 (26.4)	67 (23.8)	1	-	-	1	-	-
	(0 - 40]	2671 (73.6)	214 (76.2)	1.15	0.9-1.5	0.348	1.26	0.9-1.7	0.144
Occupation	Employed	238 (6.6)	13(4.6)	1	-	-	1	-	-
	Trader	1066 (29.3)	74 (26.3)	1.27	0.7-2.4	0.438	1.20	0.7-2.3	0.564
	Student	475 (13.1)	32 (11.4)	1.23	0.7-2.5	0.535	1.37	0.7-2.9	0.390
	Military	107 (2.9)	8 (2.8)	1.37	0.5-3.4	0.499	1.24	0.5-3.1	0.651
	Unemployed	497 (13.7)	26 (9.3)	0.96	0.5-2.0	0.901	1.20	0.5-2.8	0.659
	Others*	1249 (34.4)	128 (45.6)	1.88	1.1-3.5	0.036	1.07	0.6-2.1	0.823
Marital status	Married	2542 (70.1)	202 (72.1)	1	-	-	1	-	-
	Single	1085 (29.9)	78 (27.9)	0.90	0.7-1.2	0.468	0.72	0.5-1.0	0.057
Residence	Rural	598 (16.5)	96 (34.2)	1					
	Urban	3034 (83.5)	185 (65.8)	0.38	0.3-0.5	<0.001	0.93	0.7-1.3	0.672
Blood donor type	Voluntary	2427 (66.8)	111 (39.5)	1	-	-	1	-	-
	Family	1205 (33.2)	170 (60.5)	3.08	2.4-4.0	<0.001	1.64	1.2-2.3	**0.004**
Blood donor status	Regular	2513 (69.2)	106 (37.7)	1					
	New	1119 (30.8)	175 (62.3)	3.71	2.9-4.8	<0.001	2.91	2.1-4.0	**p<0.001**
HIV	Negative	3615 (99.5)	277 (98.6)	1	-	-	1	-	-
	Positive	17 (0.5)	4 (1.4)	3.07	0.9-8.4	0.045	2.32	0.6-6.8	0.153
HCV	Negative	3587 (98.8)	274 (97.5)	1					
	Positive	45 (1.2)	7 (2.5)	2.04	0.8-4.3	0.084	1.61	0.6-3.5	0.265
Syphilis	Negative	3502 (96.4)	258 (91.8)	1	-	-	1	-	-
	Positive	130 (3.6)	23 (8.2)	2.40	1.5-3.7	<0.001	1.38	0.8-2.2	0.199

*farmers, stockbreeders, taximen, tailors, hairdressers, Koranic masters, and transporters; OR: odds ratio; aOR: adjusted odds ratio; 95%CI: 95% confidence interval. Significant p-values are shown in **bold**.

Table 3 Risk factors associated with HCV infection in blood donors

Variables		HCV		Univariate analysis			Multivariate analysis		
		Negative	Positive	OR	95%CI	*p*-value	aOR	95%CI	*p*-value
Gender	Female	725 (188.)	10 (19.2)	1	-	-	1	-	-
	Male	3136 (81.2)	42 (80.8)	0.97	0.5-2.1	0.934	0.90	0.5-2.0	0.780
Age	(40 - 64]	2849 (73.8)	36 (69.2)	1	-	-	1	-	-
	(0 - 40]	1009 (26.2)	16 (30.8)	0.80	0.5-1.5	0.453	0.89	0.7-1.7	0.714
Marital status	Married	1150 (29.8)	13 (25.0)	1	-	-	1	-	-
	Single	2705 (70.2)	39 (75.0)	0.78	0.4-1.4	0.450	0.93	0.5-1.9	0.843
Residence	Rural	677 (17.5)	17 (32.7)	1	-	-	1	-	-
	Urban	3184 (82.5)	35 (67.3)	0.44	0.3-0.8	0.006	0.60	0.3-1.3	0.213
Blood donor type	Voluntary	1350 (35.0)	25 (48.1)	1	-	-	1	-	-
	Family	2511 (65.0)	27 (51.9)	1.72	2.0-3.0	0.052	0.97	0.4-2.1	0.945
Blood donor status	Regular	1270 (32.9)	24 (46.2)	1	-	-	1	-	-
	New	2591 (67.1)	28 (53.8)	1.75	1.0-3.0	0.046	1.20	0.6-2.4	0.601
HBV	Negative	3587 (92.9)	45 (86.5)	1	-	-	1	-	-
	Positive	274 (7.1)	7 (13.5)	2.04	0.8-4.3	0.084	1.60	0.6-3.5	0.268
Syphilis	Negative	3716 (96.2)	44 (84.6)	1	-	-	1	-	-
	Positive	145 (3.8)	8 (15.4)	4.66	2.0-9.6	<0.001	3.54	1.5-7.6	**0.002**

OR: odds ratio; aOR: adjusted odds ratio; 95%CI: 95% confidence interval. Significant p-values are shown in **bold**.

Discussion

Blood transfusion is a vital medical procedure for the recipient. However, it is associated with certain risks of transfusion-transmissible infections, including hepatitis B and C viruses. This study aimed to investigate the seroprevalence and factors associated with hepatitis B and C in CRTS-TA blood donors. This is the first-ever study to assess HBsAg and HCV seroprevalence, as well as associated risk factors, in the region of Tahoua. In this study, overall HBsAg seroprevalence was 7.2%, and HCV seroprevalence was 1.3%. These results were consistent with those obtained in the Zinder region (9.12% for HBV, 1% for HCV) (Doutchi et al., 2021). In a recent systematic review and meta-analysis, overall HBsAg seroprevalence was 6.9%, with regional disparities (Table 4) (Quintas et al., 2024).

Table 4 Frequency of HBV and HCV detection among blood donors in Niger and Africa

Country / Continent	Regions	Seroprevalence		References
		HBV	HCV	
Niger	Tahoua	7.2%	1.3%	Our study
	Zinder	9.1%	1.0%	(Doutchi et al., 2021)
	Niamey	15.4%	1.2%	(Mayaki et al., 2013)
Africa	West Africa	10.1%	-	(Quintas et al., 2024)
	East Africa	4.9%	-	(Quintas et al., 2024)
	Central Africa	7.8%	-	(Quintas et al., 2024)
	Austral Africa	2.5%	-	(Quintas et al., 2024)
	North Africa	1.7%	-	(Quintas et al., 2024)

The seroprevalence of hepatitis B virus obtained in this study was at an intermediate level according to WHO guidelines on hepatitis B virus testing, which stipulate that, about its endemicity, hepatitis B infection is classified as high when it is above 8%, as intermediate when it is between 2% and 8%, and as low endemicity when it is below 2% (OMS, 2017).

HCV and HBV seroprevalence also varied according to donor type. In this study, the seroprevalence rates were 1.1% for HCV and 4.4% for HBV among voluntary donors, while among family donors, they were 1.8% for HCV and 12.4% for HBV. These results could be explained by the fact that voluntary donors are generally regular donors who are better aware of the risk of infection. In contrast, family donors are often occasional donors who are not effectively sensitized to the prevention of viral hepatitis. Doutchi et *al*. (Doutchi et al., 2021) reported HBV seroprevalence of 9.15% in voluntary donors and 9.07% in family donors (Doutchi et al., 2021). In the same study, the HCV seroprevalence rates were 0.55% among voluntary donors and 1.69% among family donors. HBsAg and HCV seroprevalence may be underestimated in Niger. Indeed, Mayaki et *al*.(Mayaki et al., 2017) reported that the performance of the rapid tests used in Niger to screen for transfusion-transmissible infections, notably hepatitis B and C, was unsatisfactory, with 2.3% and 0.9% of results falsely negative for HBsAg and HCV, respectively (Mayaki et al., 2017).

Several risk factors were selected to understand the dynamics of the spread of hepatitis B and C among blood donors. For hepatitis B, male blood donors, family blood donors, and new blood donors were more likely to have been infected with the virus than other blood donors. These results were similar to those obtained in a study carried out among blood donors in Niamey (Mayaki et al., 2013). In Mali, authors had reported that HBV infection among blood donors was associated with low levels of education and male sex (Jary et al., 2019). This result may be due to better access to HBV vaccination for women receiving medical care during pregnancy (Jary et al., 2019). However, it could also be linked to a gender-differentiated cellular immunity, with women exhibiting greater humoral and cellular responses to antigenic stimulation, vaccination, and infection than men (Fischinger et al., 2019; Flanagan et al., 2017; Jacobsen & Klein, 2021; Li et al., 2019; Ruggieri et al., 2018; Ursin et al., 2022). For hepatitis C, syphilis-positive blood donors were significantly more likely to have been infected than other blood donors. This may be explained by the fact that hepatitis C infection and syphilis are among the STIs sharing the same mode of transmission.

There were some limitations to our study. First, the use of ELISA as a screening test instead of the rapid diagnostic test for HBsAg and HCV would have been more appropriate. Second, this study was a single-center study, and data might differ in other regions in Niger. Despite these limitations, to the best of our knowledge, this study represents the first investigation of the seroprevalence and factors associated with hepatitis B and C in CRTS-TA potential blood donors. We believe that our results provide baseline information that will further aid in controlling and preventing the transmission of hepatitis B and C viruses from blood donors in Niger.

Conclusions

This study reveals intermediate-level endemicity of HBV and low-level endemicity of HCV among blood donors in Tahoua. Male sex, family donation, and first-time donation were significantly associated with HBV infection, while syphilis co-infection was linked to HCV. These findings highlight the need for targeted donor education, improved screening protocols, and post-donation counseling to enhance transfusion safety and reduce the risk of viral transmission in Niger.

Conflict of interest: The authors declare no conflict of interest.

Ethical statement: Ethical approval was obtained from the National Health Research Ethics Committee, reference number N°04/2024/CNERS. As is customary with blood donations, all donors provided their informed consent for testing for hepatitis B and C. Ethical standards and procedures of the committee for human experimentation were adequately followed.

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Availability of data and materials: The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Authors' contributions: AY, SB, and SM: conceptualized and designed the study. AY and ZMI: conceptualized and designed the study, analyzed and curated data, and drafted the original manuscript. BAN, OT, SMS, and AS: revised and reviewed the manuscript. AO, SM, MD, and SB: coordinated and directed the study, reviewed and revised the manuscript. All authors read and approved the final manuscript

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